Calculated using the conversion factors: protein 17 kJ/g, fat 37 kJ/g, carbohydrate (available, expressed as monosaccharides) 16 kJ/g and alcohol 29 kJ/g.

Starch (STAR)

Includes dextrins but excludes resistant starch. Expressed as monosaccharide equivalents.

Oligosaccharides (OLIGO)

Expressed as monosaccharide equivalents. Any known or measured contribution from oligosaccharides and/or maltodextrins has been included in the total carbohydrate value but not in the columns for starch or total sugars. In most foods oligosaccharides are present in relatively low quantities. In vegetables and some processed foods where glucose syrups and maltodextrins are added, oligosaccharides will make a significant contribution to carbohydrate content. Where oligosaccharides are present in foods, they are not always measured separately and may be included in the starch, sugar or fibre fractions, depending on the nature of the oligosaccharide and on the analytical methods used.

Total sugars (TOTSUG)

Sugars are expressed as monosaccharide equivalents and include free monosaccharides (glucose, fructose and galactose) and disaccharides (sucrose, maltose and lactose). The value does not include any contribution from oligosaccharides present in the food.

Alcohol (ALCO)

Values are given as g/100 ml. Pure ethyl alcohol has a specific gravity of 0.79, dividing values by 0.79 converts them to alcohol by volume (ml/100 ml).

NSP (ENGFIB)

Non-starch polysaccharides (2) includes insoluble fibre (cellulose, insoluble non-cellulosic polysaccharides) and soluble fibre (soluble cellulosic polysaccharides).

AOAC fibre (AOACFIB)

AOAC determinations (3) include resistant starch and lignin in the estimation of total fibre, rather than only the non-starch polysaccharides.

Fatty acids

Values for total saturated (SATFOD), monounsaturated (MONOFOD), polyunsaturated (POLYFOD) and trans fatty acids (FODTRANS) are given as well as values for branched chain saturated fatty acids (TOTBRFOD), cis-monounsaturated (MONOFODc) and cis-polyunsaturated fatty acids (POLYFODc). Trans fatty acids are also included in total monounsaturated and total polyunsaturated fatty acids. For food labelling purposes trans fats are not included in the values for monounsaturated and polyunsaturated fats. Values for total fatty acids are expressed as both g/100g food and also g/100g fatty acid methyl esters. The fat in most foods contains non-fatty acid material such as phospholipids and

sterols and to allow the calculation of the total fatty acids in a given weight of food, the glycerol conversion factors given in the 'Factors' worksheet were applied.

Cholesterol (CHOL)

Values are expressed as mg/100g food. To convert to mmol cholesterol, divide the values by 386.6.

Inorganics worksheet

Values for are given for:

Sodium (NA)
Potassium (K)
Calcium (CA)
Magnesium (MG)
Phosphorus (P)
Iron (FE)
Copper (CU)
Zinc (ZN)
Chloride (CL)
Selenium (SE)
Iodine (I)

Vitamins worksheet

Fat-soluble vitamins

The 2 components of vitamin A are given separately as Retinol (RET) and Carotene (CAREQU).

Retinol (RET)

Retinol (RET) is expressed as the weight of all-trans-retinol equivalent, that is, the sum of all-trans-retinol plus contributions from the other forms after correction to account for their relative activities (4). Where the retinol profile was incomplete, because values for 13-cis retinol and/or retinaldehyde were not available, it has been assumed that only all-trans retinol is present, leading to a possible underestimate in some cases.

Carotene (CAREQU)

Represents the β-carotene activity and is the sum of the β-carotene and half of any α-carotene or cryptoxanthins present. Where the carotenoid profile was incomplete, because only values for β-carotene were available, it has been assumed that only β-carotene is present. This may result in an underestimate of β-carotene equivalents, but as α-carotene and cryptoxanthin are usually present in low levels in foods without complete carotenoid profiles, it is likely that any error is small.

Total retinol equivalent (RETEQU)

The generally accepted relationship is that 6 µg β-carotene or 12 µg of other active carotenoids are equivalent to 1 µg of retinol (5), that is:

$$\text{Vitamin A potency as retinol equivalent} = \mu g \text{ retinol} + \frac{\mu g \text{ β-carotene equivalent}}{6}$$

Vitamin D (VITD)

Few foods contain vitamin D from intrinsic sources. All those which contain vitamin D naturally are products of animal origin and contain D_3 (cholecalciferol) derived, as in humans, from the action of sunlight on the animal's skin or from its own food. Vitamin D_2 (ergocalciferol) made commercially has the same potency as D_3 in man. Vitamin D_2 and vitamin D_3 are both used to fortify a number of foods.

Meat can contain vitamin D_3 (cholecalciferol) derived from the action of sunlight or, for pigs and poultry, from the feed. Vitamin D_3 in meat may also be present in the form of the more active 25-hydroxy vitamin D3. For meat, meat products, and poultry, therefore, the total vitamin D activity has been taken as the sum of vitamin D_3 (cholecalciferol) and 5 times 25-hydroxy vitamin D_3 (25-hydroxy cholecalciferol), where data are available.

Vitamin E (α-tocopherol equivalent) (VITE)

Values take into account vitamin E activity using conversion factors (6) and are expressed as α-tocopherol equivalents.

Vitamin K_1 (VITK1)

Phylloquinone, the predominant, naturally-occurring, vitamin K in foods.

Water-soluble vitamins

Thiamin (THIA)

Values are expressed as thiamine chloride hydrochloride.

Niacin (NIAC)

Values are the sum of nicotinic acid and nicotinamide.

Tryptophan/60 (TRYP60)

Potential nicotinic acid from the amino acid tryptophan, calculated as tryptophan divided by 60.

Vitamin B_6 (VITB6)

Values are expressed as pyridoxine hydrochloride by microbiological assay, or the sum of the individual forms by HPLC, and expressed as the sum of the total pyridoxine hydrochloride, pyridoxal hydrochloride and pyridoxamine dihydrochloride.

Folate (FOLT)
Values are expressed as total folates measured after deconjugation of the polyglutamyl forms.

Pantothenate (PANTO)
Values are expressed as calcium D-pantothenate.

Vitamin C (VITC)
Values include ascorbic acid and dehydroascorbic acids.

Vitamin fractions worksheet

Values are given for some foods for vitamers that contribute to retinol equivalents (all-trans retinol, 13-cis-retinol, dehydroretinol and retinaldehyde), carotene equivalents (alpha-carotene, beta-carotene and beta-cryptoxanthins), vitamin D (cholecalciferol and 25-hydroxy-vitamin D_3) and vitamin E equivalents (alpha, beta, gamma and delta tocopherol and alpha, beta, gamma and delta tocotrienols). In addition, there are some values for lutein, lycopene and 5-methyl folate.

FATTY ACID worksheets

Worksheets are given containing values for individual fatty acid isomers where these are available. There are worksheets for saturated fatty acids, monounsaturated fatty acids and polyunsaturated fatty acids. Values for total fatty acids are expressed as both g/100g food and also g/100g fatty acid methyl esters. The fat in most foods contains non-fatty acid material such as phospholipids and sterols and to allow the calculation of the total fatty acids in a given weight of food, the glycerol conversion factors given in the 'Factors' worksheet were applied.

PHYTOSTEROL worksheet

Values are given for some foods for a range of phytosterols including beta-sitosterol, brassicasterol, campesterol, delta-5-avenasterol, delta-7-avenasterol, delta-7-stigmastenol and stigmasterol.

ORGANIC ACIDS worksheet

Values are given for a few foods for citric acid and malic acid.

Food labelling

Nutrition information is increasingly being provided on food labels and the nutrition declaration ('back of pack' labelling) has been mandatory on the majority of pre-packed foods since 2016. Values from CoFID may be used to calculate label data.

Food information regulations retained Regulation (EU) No. 1169/2011 brought rules on general and nutrition labelling together into a single regulation (replacing the previous food labelling regulations). Under the regulations (available at: retained Regulation (EU) No 1169/2011, 'back of pack' nutrition labelling became mandatory for the majority of pre-packed foods from 13 December 2016.

The mandatory declaration comprises:

energy (kJ, kcal)
fat
saturates
carbohydrate
sugars
protein
salt

Salt is calculated as total sodium content multiplied by 2.5. Supplementary information on other nutrients listed in the Regulation can be provided on a voluntary basis. The additional listed nutrients are: monounsaturates; polyunsaturates; polyols; starch; fibre; and specified minerals and vitamins, present in significant amounts (as defined in the Regulation). If a claim is made for any of these nutrients, or if minerals and/or vitamins are added to a food, then the amount of the respective nutrient(s) must be declared in addition to the mandatory declaration outlined above.

Declared values for nutrients should be average values derived using one or more of the following methods:

- manufacturer's analysis of food
- a calculation from the known or actual average values of the ingredients used in the preparation of the food
- a calculation from generally established and accepted data.

Generally established and accepted data for the UK include values published in CoFID, if the product or its ingredients are similar to those described. Nevertheless, it is important to note the following differences:

- protein should be given as total nitrogen x 6.25 for every food, whereas more specific factors have been used in CoFID
- carbohydrate is to be declared as the weight of the carbohydrates themselves and not their monosaccharide equivalents.

The following factors may be used to convert monosaccharide equivalents from this edition to actual weights:

Total carbohydrate	**Divide by 1.05 (unless it is known to be mainly starch or mainly oligosaccharide)**
Starch	Divide by 1.10
Sucrose and lactose	Divide by 1.05
Glucose, etc.	As given

Different factors are to be used to calculate energy values and are shown below:

	kcal/g	**kJ/g**
Carbohydrate (except polyols), expressed as weight	4	17
Polyols	2.4	10
Protein	4	17
Fat	9	37
Salatrims	6	25
Alcohol (ethanol)	7	29
Organic acid	3	13
Fibre	2	8
Erythritol	0	0

'Front of pack' nutrition labelling

Retained Regulation (EU) No. 1169/2011 allows elements of the mandatory nutrition declaration which are of importance to public health to be repeated on the 'front of pack' in one of the following formats:

- energy value alone
- energy value plus amounts of fat, saturates, sugars and salt.

You can read guidance on providing 'front of pack' labelling in line with UK government 2013 recommendation.

Tolerances for nutrient values declared on a label

It is widely recognised that it is not possible for foods to always contain the exact quantity of nutrients declared on the label, owing to natural variation, and variations during food production and storage. However, in order to avoid consumers being misled, it is important that the deviation from declared values should be minimal. The Department of Heath and Social Care have produced a Technical guidance on nutrition labelling which contains information on tolerances, that is, the acceptable differences between the nutrient values declared on a label and those established in the course of official controls by enforcement authorities.This content is based on a guidance document produced by the European Commission which contains additional information.

Appendices

Appendix A: Main data references

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